EE 102: Signal Processing and Linear Systems

Problem Set #8: Practice and Study Guide for Midterm Exam #2

Name: _____ **Submission Date:** _____

Please note the following:

1. This practice guide is longer than the exam.
2. Midterm Exam #2 on Wednesday, October 28 (1:40–2:55 PM). Problem set is due **Wednesday, October 28, at 11:59 PM**, after the exam.
3. The format of questions on the exam should not be gleaned from the problem set. The study guide is meant to provide practice and review of the concepts, not to predict the exam questions.

Use the following Fourier transform convention

$$X(\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t}\,dt, \qquad x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(\omega)e^{j\omega t}\,d\omega.$$

Problem 1 Frequency content of signals.

Frequency content of signals is often the property that engineers look for when analyzing systems. In this problem, we will analyze the frequency content of signals.

(a) [5 points] A signal is written as

$$z(t) = 3e^{j2t} + 3e^{-j2t}.$$

Express it using sinusoids, identify its amplitude, fundamental angular frequency, and describe its frequency content.

Hint: $e^{j\theta} = \cos\theta + j\sin\theta$.

(b) [5 points] Sketch

$$p(t) = u(t) - u(t-1), \qquad q(t) = 1 + \cos(2t).$$

What are the frequency contents of each of the signals? A descriptive answer is fine.

Problem 2 Application example.

Consider a series RC circuit with $R = 2\,\mathrm{k}\Omega$ and $C = 250\,\mu\mathrm{F}$. Measure the output voltage across the capacitor. Assume zero initial capacitor voltage. You may use

$$\tau_C = RC, \qquad h_C(t) = \frac{1}{\tau_C} e^{-t/\tau_C} u(t), \qquad y(t) = \int_{-\infty}^{\infty} x(\lambda) h_C(t-\lambda)\, d\lambda.$$

(a) [5 points] Calculate τ_C and write $h_C(t)$ with numerical constants. Find the output for $x(t) = 3\delta(t - \frac{1}{2})$. When does the response begin?

Hint: $x(t) * A\delta(t - t_0) = Ax(t - t_0)$.

(b) [10 points] For the same circuit with $x(t) = 6u(t)$ volts, evaluate the convolution integral to find the capacitor voltage for all t. Find the voltage at $t = 0^+$, $t = 0.5$, $t = 1$, and as $t \to \infty$.

Problem 3 Practice Fourier Series.

(a) [5 points] For

$$r(t) = 1 + 4\cos(6t) - 2\sin(10t),$$

find the fundamental angular frequency and period, and list every nonzero complex Fourier coefficient, including the DC term.

Hints: Find a common fundamental frequency for 6 and 10 rad/s. Then use

$$\cos\theta = \frac{e^{j\theta} + e^{-j\theta}}{2}, \qquad \sin\theta = \frac{e^{j\theta} - e^{-j\theta}}{2j}.$$

(b) [5 points] A signal generator produces the periodic voltage $v(t+3) = v(t)$. Over $-3/2 \le t < 3/2$, define

$$v(t) = \begin{cases} 6, & |t| < 1/2, \\ 0, & 1/2 < |t| < 3/2. \end{cases}$$

Sketch two periods. Find T, f_0, and ω_0, and identify what fraction of each period the voltage is nonzero.

Hint: The pulse has width 1 second and repeats every 3 seconds. Break down the integral accordingly.

(c) [10 points] For the signal in (b), derive a_0 and a_k for $k \neq 0$ using the Fourier series analysis equation. Identify the nonzero indices for which $a_k = 0$ and explain the meaning of a_0 for this voltage.

$$a_k = \frac{1}{T}\int_{-T/2}^{T/2} v(t)e^{-jk\omega_0 t}\,dt.$$

Problem 4 Energy content of signals.

A signal generator can produce the following voltages, all with period $T = 2$ seconds and peak magnitude 3:

$$s(t) = 3\cos(\pi t), \qquad b(t) = \begin{cases} 3, & 0 \le t < 1, \\ -3, & 1 \le t < 2, \end{cases} \qquad b(t+2) = b(t),$$

and the sawtooth $r(t) = 3t$ for $-1 \le t < 1$, with $r(t+2) = r(t)$. For this problem, the coefficients are supplied so you can concentrate on their energy interpretation:

$$\begin{aligned} s: &\quad c_1 = c_{-1} = \tfrac{3}{2}, \quad c_k = 0 \text{ otherwise}, \\ b: &\quad d_0 = 0, \quad d_k = \begin{cases} \dfrac{6}{j\pi k}, & k \text{ odd}, \\ 0, & k \neq 0 \text{ even}, \end{cases} \\ r: &\quad e_0 = 0, \quad e_k = \frac{3j(-1)^k}{\pi k} \quad (k \neq 0). \end{aligned}$$

(a) [5 points] Use Parseval's theorem to calculate the average energy over a period (power) of each voltage. Give numerical exact values.

$$P_x = \frac{1}{T}\int_{t_0}^{t_0+T} |x(t)|^2\, dt = \sum_{k=-\infty}^{\infty} |a_k|^2.$$

Hints: Include both signs of the index. You may use

$$\sum_{k=1}^{\infty} \frac{1}{k^2} = \frac{\pi^2}{6}, \qquad \sum_{m=0}^{\infty} \frac{1}{(2m+1)^2} = \frac{\pi^2}{8}.$$

(b) [3 points] Assuming other things are same, rank the three voltages by the average energy over a period. Then calculate the energy over one period for each.

Hint: $E_{x,T} = TP_x$. For this part, you may use $P_s = 9/2$, $P_b = 9$, and $P_r = 3$ independently of the preceding sum.

(c) [4 points] For each waveform, calculate the fundamental pair's contribution $P_{\text{fund}} = |a_1|^2 + |a_{-1}|^2$ to P_x, and the fraction P_{fund}/P_x. Use the corresponding supplied coefficients c_k, d_k, or e_k in place of a_k.

(d) [3 points] Which waveform has the greatest total average energy over a period? Which has the greatest fraction in its fundamental? Use these results to explain why the waveform with the most total energy need not be the closest to a single sinusoid.

Problem 5 Standard Fourier transforms.

(a) [10 points] A voltage pulse is

$$p(t) = 3[u(t+1) - u(t-1)].$$

Sketch it and derive $P(\omega)$ directly from the transform integral. Find $P(0)$ and the first positive zero. Evaluate $P(\pi/2)$. Explain what the zero-frequency value tells you about the pulse.

Hints: Integrate over $-1 < t < 1$. You may write your answer using $\text{sinc}(z) = \sin z/z$, with $\text{sinc}(0) = 1$. At $\omega = 0$, use the signed area of the pulse.

(b) [5 points] Find the transform of

$$q(t) = 2\delta(t-1) - \delta(t+1).$$

Evaluate $Q(0)$ and $Q(\pi/2)$. Give their magnitudes and phases, and explain how the shifted impulses affect the frequency-domain result.

Hint: Sifting gives $\int f(t)\delta(t-t_0)\,dt = f(t_0)$, with the integral over all time. You may use $e^{\pm j\pi/2} = \pm j$.

(c) [5 points] Use the inverse transform integral and sifting to reconstruct the real signal corresponding to

$$X(\omega) = 4\pi\delta(\omega) + \pi\delta(\omega-2) + \pi\delta(\omega+2).$$

Hint: Each impulse contributes its weight times $e^{j\omega t}/(2\pi)$ at the impulse's frequency. Combine the two nonzero-frequency terms using Euler's formula.

Problem 6 RC spectra and frequency content

A voltage-measurement circuit uses a series RC network with output across the capacitor. Here $R = 1\,\mathrm{k}\Omega$, $C = 250\,\mu\mathrm{F}$, and the initially relaxed circuit has impulse response

$$h(t) = 4e^{-4t}u(t).$$

This circuit's numerical values are independent of Problem 2.

(a) [6 points] Derive $H(\omega)$ by applying the Fourier transform to $h(t)$. Find $H(0)$ and the magnitude and phase at $\omega = 4$. Describe what happens to the magnitude as $|\omega|$ grows.

Hints: Integrate $4e^{-(4+j\omega)t}$ over $0 \le t < \infty$. For $a > 0$, $|a+j\omega| = \sqrt{a^2+\omega^2}$ and $\arg(a+j\omega) = \tan^{-1}(\omega/a)$.

(b) [6 points] The test input consists of two impulses:

$$x(t) = \delta(t) + \delta(t - \tfrac{1}{2}).$$

Find the output $y(t)$ using LTI properties. Then obtain $Y(\omega)$ from the output signal using linearity and a change of variables in the delayed term. Show why the delay contributes a complex exponential.

Hints: Use $y(t) = h(t) + h(t - \frac{1}{2})$. For this and the remaining parts, you may use $H(\omega) = 4/(4 + j\omega)$ independently of part (a). In the delayed transform integral, set $\lambda = t - \frac{1}{2}$.

(c) [4 points] A spectrum display for the two-impulse test is described by

$$Y(\omega) = \frac{4}{4 + j\omega}\left(1 + e^{-j\omega/2}\right).$$

Using this supplied expression independently of your earlier answers, find $Y(0)$ and $Y(2\pi)$. The circuit has $H(2\pi) \neq 0$. Explain why the output nevertheless has a zero at 2π rad/s. What cancels at this frequency?

Hint: Evaluate the factor in parentheses using $e^{-j\pi} = -1$.

(d) [4 points] A second version of the circuit uses the same $R = 1\,\mathrm{k}\Omega$ but doubles the capacitor to $500\,\mu\mathrm{F}$. For each version, calculate RC, the DC gain, and the magnitude of the frequency response at $\omega = 4$. Which version reduces a component at 4 rad/s more? Explain what happens to the charging timescale when the capacitance doubles.

Hints: For a capacitor-voltage RC circuit,

$$H(\omega) = \frac{1}{1 + j\omega RC}, \qquad \tau_C = RC.$$

You may use $1/\sqrt{2} \approx 0.707$ and $1/\sqrt{5} \approx 0.447$.